The first law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a consistent sign convention in an energy balance.

SUCCESS CRITERIA - I can...

1. State whether heat enters or leaves.
2. Distinguish work done by the system from work done on it.
3. Apply ΔU=Q-W with the stated signs.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “internal energy” and “work” to explain your model or result. Sentence starters are optional.

internal energy: energy stored in a system’s microscopic motion/interactions

work: energy transferred by a force along a displacement

surroundings: everything outside the chosen system

01 Warm-up: bring back an earlier idea

A 0.3 kg vapor releases 100000 J while condensing at constant temperature.
Find latent heat and signed Q for the sample.

Word help: latent heat = energy per mass needed for a specified phase change

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta U = Q - W_{by}$.
Q is positive into the system; W_{by} is positive when the system does work on its surroundings.

Worked reasoning: A system receives 90 J of heat and does 30 J of work on its surroundings. With ΔU=Q-W, internal energy increases by 60 J.

Your turn: Draw labeled arrows crossing a system boundary and mark the 60 J stored increase.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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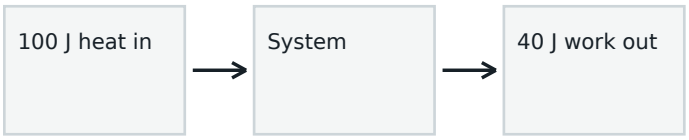
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A system absorbs 100 J of heat and does 40 J of work.
Draw the energy arrows and find its internal-energy change.

Sentence starter: I found ... [unit]. This value means ... in this situation.



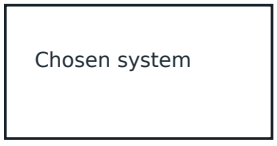
Marzano check - circle: ① ② ③

04 Apply the physics

A gas loses 50 J of heat while a piston does 60 J of work on it.
Label both transfers on the system boundary.
Use $\Delta U = Q - W_{by}$, with work done by the gas positive, to calculate ΔU.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

05 Explain your reasoning

Why does the phrase “work is 60 J” need a direction or sign convention?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A 4 kg cart at 12 m/s hits an identical cart at rest. Final velocities are 5 and 7 m/s.
Check momentum and energy; classify the collision.
Show the energy changes and transfers; label units.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: The ... in my model represents ... . Its label shows ... .



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

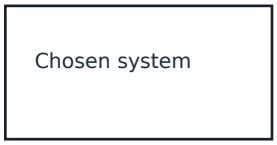
Sentence starters are optional. You may plan in any language.

07 Exit check

A process has ΔU=-15 J and W_{by}=25 J.
Find Q and say whether heat enters or leaves.
Label energy entering, leaving or stored on the system model.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

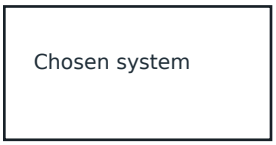
Marzano check - circle: 1 2 3

08 STRETCH: test the model

A system loses 20 J of heat and has 50 J of work done on it. Using work-by-system as positive, find Q, W and ΔU. Then reverse both transfers.
Label energy entering, leaving or stored on the system model.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3